

Single CRUD

```
package MONGO;

import com.mongodb.client.*;
import com.mongodb.client.model.Filters;
import com.mongodb.client.model.Updates;
import org.bson.Document;

public class MongoCRUExample {

    public static void main(String[] args) {
        // Connect to MongoDB
        MongoClient mongoClient = MongoClient.create("mongodb://localhost:27017");
        // Access database
        MongoDBDatabase database = mongoClient.getDatabase("CollegeDB");
        // Access collection
        MongoCollection<Document> collection =
            database.getCollection("Students");
        // =====
        // CREATE (Insert)
        // =====
        Document student = new Document("id", 101)
            .append("name", "John")
            .append("department", "CSE")
            .append("cgpa", 8.7);
        collection.insertOne(student);
        System.out.println("Document Inserted Successfully");
        // =====
        // READ (Find)
```

```

// =====
System.out.println("\nStudent Records:");
FindIterable<Document> students = collection.find();
for (Document doc : students) {
    System.out.println(doc.toJson());
}
// =====
// UPDATE
// =====
collection.updateOne(
    Filters.eq("id", 101),
    Updates.set("cgpa", 9.2)
);
System.out.println("\nDocument Updated Successfully");
// =====
// READ Updated Record
// =====
Document updatedStudent =
collection.find(Filters.eq("id", 101)).first();
System.out.println("\nUpdated Record:");
System.out.println(updatedStudent.toJson());
// =====
// DELETE
// =====
collection.deleteOne(Filters.eq("id", 101));
System.out.println("\nDocument Deleted Successfully");
// Close connection
mongoClient.close();
}

```

}

```
Problems Javadoc Declaration Console
<terminated> MongoCRUDEExample (1) [Java Application] C:\Program Files\Java\jre1.8.0_481\bin\javaw.exe (Jun 3, 2026, 6:37:36 PM)
Jun 03, 2026 6:37:36 PM com.mongodb.internal.diagnostics.logging.Loggers shouldUseSLF4J
WARNING: SLF4J not found on the classpath. Logging is disabled for the 'org.mongodb.driver' component
Document Inserted Successfully

Student Records:
{"_id": {"$oid": "6a2027181af871969d3ca012"}, "id": 101, "name": "John", "department": "CSE", "cgpa": 8.7}

Document Updated Successfully

Updated Record:
{"_id": {"$oid": "6a2027181af871969d3ca012"}, "id": 101, "name": "John", "department": "CSE", "cgpa": 9.2}

Document Deleted Successfully
```

MULTIPLE CRUD

package MONGO;

import com.mongodb.client.*;

import com.mongodb.client.model.Filters;

import com.mongodb.client.model.Updates;

import org.bson.Document;

public class MongoCRUDEExample2 {

public static void main(String[] args) {

MongoClient mongoClient =

MongoClients.create("mongodb://localhost:27017");

MongoDatabase database =

```
mongoClient.getDatabase("CollegeDB");
```

```
MongoCollection<Document> collection =  
    database.getCollection("Students");
```

```
// INSERT MULTIPLE DOCUMENTS
```

```
collection.insertOne(  
    new Document("id", 101)  
    .append("name", "John")  
    .append("department", "CSE")  
    .append("cgpa", 8.7));
```

```
collection.insertOne(  
    new Document("id", 102)  
    .append("name", "Alice")  
    .append("department", "IT")  
    .append("cgpa", 9.1));
```

```
collection.insertOne(  
    new Document("id", 103)  
    .append("name", "Bob")  
    .append("department", "ECE")  
    .append("cgpa", 8.4));
```

```
System.out.println("3 Documents Inserted");
```

```
// DISPLAY ALL RECORDS
```

```
System.out.println("\nStudents Before Update:");
```

```
FindIterable<Document> students = collection.find();
```

```
for (Document doc : students) {  
    System.out.println(doc.toJson());  
}
```

```
// UPDATE MULTIPLE RECORDS
```

```
collection.updateOne(  
    Filters.eq("id", 101),  
    Updates.set("cgpa", 9.0));
```

```
collection.updateOne(  
    Filters.eq("id", 102),  
    Updates.set("department", "CSE"));
```

```
collection.updateOne(  
    Filters.eq("id", 103),  
    Updates.set("name", "Robert"));
```

```
System.out.println("\n3 Documents Updated");
```

```
// DISPLAY UPDATED RECORDS
```

```
System.out.println("\nStudents After Update:");
```

```
students = collection.find();
```

```
for (Document doc : students) {  
    System.out.println(doc.toJson());  
}  
  
// DELETE MULTIPLE RECORDS  
  
collection.deleteOne(Filters.eq("id", 101));  
  
collection.deleteOne(Filters.eq("id", 102));  
  
System.out.println("\n2 Documents Deleted");  
  
// DISPLAY FINAL RECORDS  
  
System.out.println("\nStudents After Delete:");  
  
students = collection.find();  
  
for (Document doc : students) {  
    System.out.println(doc.toJson());  
}  
  
mongoClient.close();  
}  
}
```

```
Problems Javadoc Declaration Console
<terminated> MongoCRUDExample2 [Java Application] C:\Program Files\Java\jre1.8.0_481\bin\javaw.exe (Jun 3, 2026, 7:17:35 PM)
Jun 03, 2026 7:17:35 PM com.mongodb.internal.diagnostics.logging.Loggers shouldUseSLF4J
WARNING: SLF4J not found on the classpath. Logging is disabled for the 'org.mongodb.driver' component
3 Documents Inserted

Students Before Update:
{"_id": {"$oid": "6a203078e647adbfe4aeb000"}, "id": 101, "name": "John", "department": "CSE", "cgpa": 8.7}
{"_id": {"$oid": "6a203078e647adbfe4aeb001"}, "id": 102, "name": "Alice", "department": "IT", "cgpa": 9.1}
{"_id": {"$oid": "6a203078e647adbfe4aeb002"}, "id": 103, "name": "Bob", "department": "ECE", "cgpa": 8.4}

3 Documents Updated

Students After Update:
{"_id": {"$oid": "6a203078e647adbfe4aeb000"}, "id": 101, "name": "John", "department": "CSE", "cgpa": 9.0}
{"_id": {"$oid": "6a203078e647adbfe4aeb001"}, "id": 102, "name": "Alice", "department": "CSE", "cgpa": 9.1}
{"_id": {"$oid": "6a203078e647adbfe4aeb002"}, "id": 103, "name": "Robert", "department": "ECE", "cgpa": 8.4}

2 Documents Deleted

Students After Delete:
{"_id": {"$oid": "6a203078e647adbfe4aeb002"}, "id": 103, "name": "Robert", "department": "ECE", "cgpa": 8.4}
```